HDOC Day-43 Assignment

Question

Write a menu-driven C program to perform the following operations using separate functions:

- Find the longest word in a sentence.
- Check if one string is a rotation of another.

1. Problem Understanding

The program is menu-driven and performs two string operations using separate functions. It can find the longest word in a sentence and check whether one string is a rotation of another.

2. Standard Test Cases

S.No.	All Inputs	Expected Output
1	Choice = 1 Sentence: C programming is very interesting	Longest word: programming
2	Choice = 2 String 1: ABCD String 2: CDAB	The strings are rotations.
3	Choice = 2 String 1: hello String 2: olelh	The strings are not rotations.

3. Algorithm

1. Start.
2. Display the menu.
3. Read the user choice.
4. For choice 1, read a sentence and call longestWord().
5. Compare the lengths of the words and store the longest word.
6. Display the longest word.
7. For choice 2, read two strings and call isRotation().
8. Check that both strings have equal length.
9. Concatenate the first string with itself and search for the second string.
10. Display the result.
11. Stop.

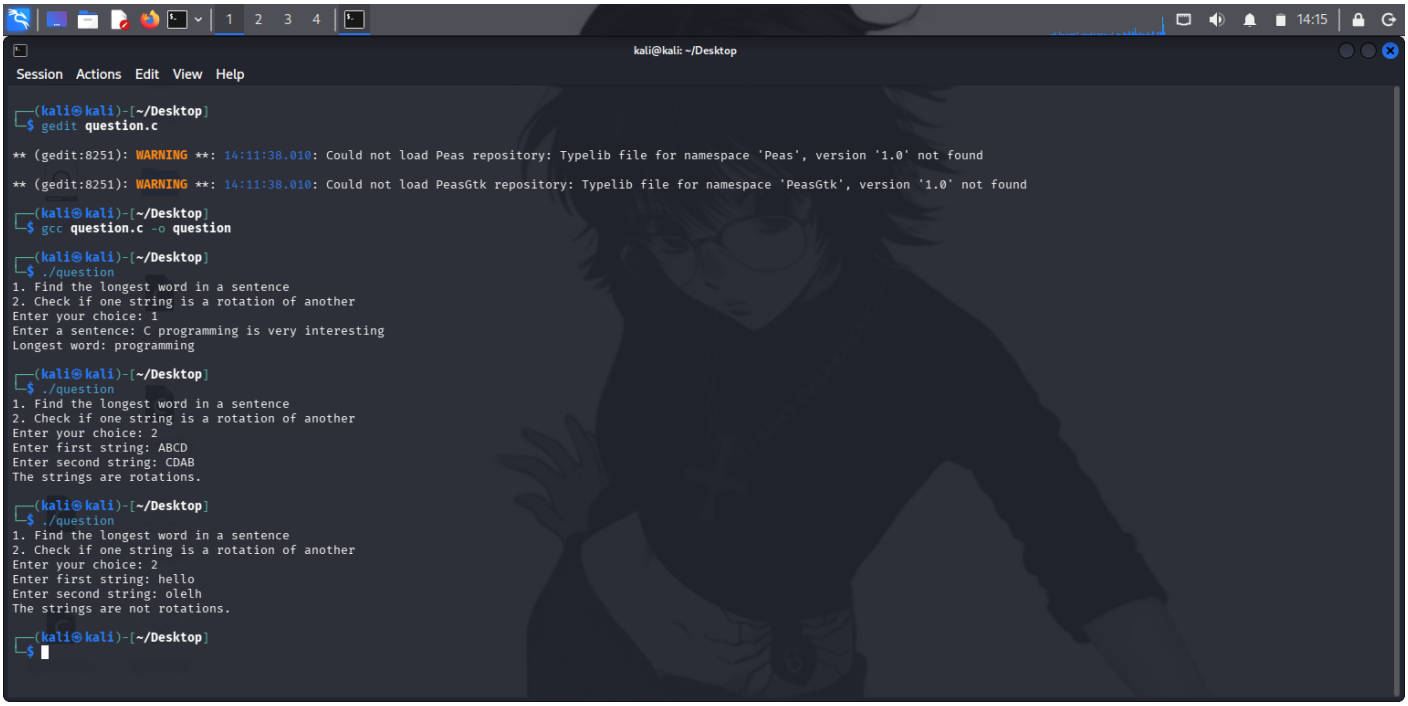
4. C Program

```
#include <stdio.h>
#include <string.h>
void longestWord(char sentence[])
{
  char word[100], longest[100] = "";
  int i = 0, j = 0;
```

```
    while (1)
    {
        if (sentence[i] != ' ' && sentence[i] != '\0' && sentence[i] != '\n')
            word[j++] = sentence[i];
        else
        {
            word[j] = '\0';
            if (j > strlen(longest))
                strcpy(longest, word);
            j = 0;
            if (sentence[i] == '\0' || sentence[i] == '\n')
                break;
        }
        i++;
    }
    printf("Longest word: %s\n", longest);
}
int isRotation(char str1[], char str2[])
{
    char temp[200];
    if (strlen(str1) != strlen(str2))
        return 0;
    strcpy(temp, str1);
    strcat(temp, str1);
    return strstr(temp, str2) != NULL;
}
int main()
{
    int choice;
    char sentence[200], str1[100], str2[100];
    printf("1. Find the longest word in a sentence\n");
    printf("2. Check if one string is a rotation of another\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
```

```
    getchar();
    if (choice == 1)
    {
        printf("Enter a sentence: ");
        fgets(sentence, sizeof(sentence), stdin);
        longestWord(sentence);
    }
    else if (choice == 2)
    {
        printf("Enter first string: ");
        fgets(str1, sizeof(str1), stdin);
        str1[strcspn(str1, "\n")] = '\0';
        printf("Enter second string: ");
        fgets(str2, sizeof(str2), stdin);
        str2[strcspn(str2, "\n")] = '\0';
        if (isRotation(str1, str2))
            printf("The strings are rotations.\n");
        else
            printf("The strings are not rotations.\n");
    }
    else
        printf("Invalid choice.\n");
    return 0;
}
```

5. Compilation and Execution Evidence



6. Verification of All Test Cases

Test	Input Summary	Actual Output	Result
1	C programming is very interesting	Longest word: programming	Pass
2	ABCD, CDAB	The strings are rotations.	Pass
3	hello, olelh	The strings are not rotations.	Pass